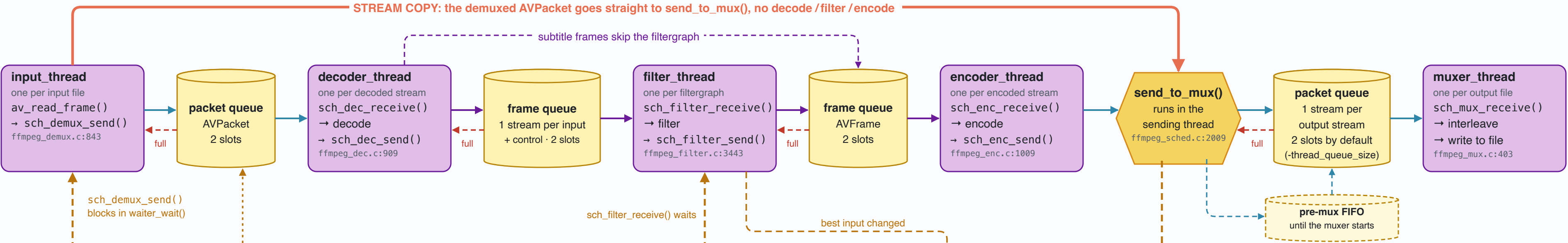


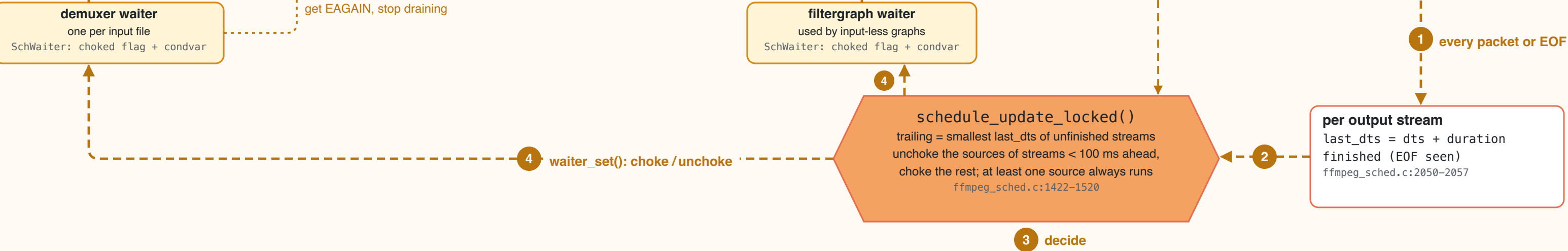
FFmpeg CLI: threaded transcoding and the Scheduler that keeps outputs in step

Each component runs in its own thread and talks only to Scheduler-owned queues. The Scheduler is not a thread: it watches output progress and blocks sources that run ahead.

MEDIA DATA PLANE packets and frames move left to right; every queue is a bounded ThreadQueue owned by the Scheduler



SCHEDULER one shared object, not a thread: its functions run inside whichever thread calls them, under schedule_lock



→ AVPacket → AVFrame → stream copy - - - → queue full: the sender blocks in tq_send() - - - → Scheduler: progress in, choke / unchoke out

Threads start in sch_start(): encoders, filtergraphs, decoders, then demuxers; a muxer starts once all its streams are ready. Optional SyncQueue and codec-internal threads not shown.

Source: FFmpeg fftools @ a344f09 · ffmpeg_sched.c, thread_queue.c, ffmpeg_{demux,dec,filter,enc,mux}.c